Swirl-String-Theory_Canon-v0.8.0

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Abstract

This document presents Swirl-String Theory (SST) as a structured canonical synthesis rather than as a claim that every phenomenological sector has already been derived to theorem level. The text consolidates the primitive constant chain linking the effective fluid density, characteristic swirl speed, Compton anchoring, and core scale; records delay-induced mode selection as the preferred non-operator route to spectral discreteness; and organizes the atomic bridge, spectroscopic filter, relational-time sector, and topology-based mass program within one explicitly stratified framework. Throughout, orthodox inputs, SST-internal derivations, bridge-level constructions, and conjectural extensions are kept distinct so that the canon remains internally coherent, externally comparable, and open to falsification.

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1 Philosophical Prelude

1.1 Why a philosophical prelude is necessary

The SST canon is not only a collection of equations but also a statement about what counts as fundamental. Standard quantum theory and general relativity are highly successful formalisms, yet they begin from different ontological primitives: operator-valued states in one case and spacetime geometry in the other. SST instead adopts a single material substrate as the underlying level of description and treats particles, radiation, and clock structure as organized states of that substrate.

1.2 The ontological shift

The canon therefore makes the following conceptual replacement:

$$\begin{aligned} \text{point particles} &\longrightarrow \text{stable swirl-string configurations,} && (1)\\ \text{vacuum as background} &\longrightarrow \text{vacuum as structured medium,} && (2)\\ \text{quantization as postulate} &\longrightarrow \text{discreteness from dynamics and topology.} && (3) \end{aligned}$$

This is not a denial of effective particle language. Rather, it is a claim that particle language is secondary and emerges from a more primitive hydrodynamic and topological layer.

1.3 Relation to orthodox physics

SST is not introduced as a replacement for the empirical content of orthodox physics. Its canonical role is instead threefold:

1. to reproduce known low-energy structures in the appropriate limits,
2. to identify which apparently fundamental constants or rules may be structurally redundant,
3. to supply a dynamical mechanism for phenomena that are otherwise postulated axiomatically.

In this sense the canon follows a *Rosetta principle*: whenever a standard description already works, SST must map to it before claiming any deeper reinterpretation.

1.4 Why delay and topology matter

Two themes recur across the development of the canon. First, finite circulation delay on closed carriers provides a natural source of temporal nonlocality. Second, topological invariants constrain admissible configurations and stabilize structures that would otherwise collapse or disperse. Together they motivate the canonical SST slogan

delay selects modes, topology protects them.

This is the philosophical reason why the canon treats quantization neither as a primitive operator rule nor as a purely statistical statement.

1.5 Scope of the present canon

The present version of the canon should be read as a structured reference document rather than a claim that every sector has already been derived to theorem level. Some components are mature and reusable, such as the primitive constant chain, circulation quantization, and the Swirl-Clock. Other components are deliberately retained as research-track layers, such as full gauge emergence, topology-to-charge assignments, and complete many-body atomic closure. The point of the canon is therefore not maximal rhetorical closure, but explicit logical organization.

1.6 Interpretive summary

The philosophical stance of SST can be summarized in one sentence:

Physics is interpreted as the dynamics of a continuous medium whose stable, delayed, and topologically organized motions appear effectively as particles, clocks, fields, and spectra.

2 Formal Foundations

2.1 Primitive Structure of the Theory

We define the Swirl-String Theory (SST) as a continuum-based framework built on a minimal set of primitive quantities.

Primitive set:

$$\mathcal{P} = \{\rho_f, \mathbf{v}_{\circlearrowleft}, \omega_c\} \tag{4}$$

where:

- ρ_f is the effective background fluid density,
- $\mathbf{v}_{\circlearrowleft}$ is the characteristic swirl velocity,
- ω_c is the Compton angular frequency.

[ORTHODOX]: $\omega_c = \frac{m_e c^2}{\hbar}$ is a standard physical constant.

[SPECULATIVE]: The identification of ω_c as a primitive dynamical frequency of the vortex core is an SST postulate.

2.2 Derived Length Scale

The characteristic core radius is defined as:

$$r_c = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{\omega_c} \tag{5}$$

[SPECULATIVE] Definition: We treat Eq. (5) as a canonical closure relation linking the characteristic core scale to the primitive pair $(\|\mathbf{v}_{\circlearrowleft}\|, \omega_c)$.

[DERIVED] Consequence: This removes r_c as an independent parameter and eliminates one source of circular parameterization.

2.3 Circulation Quantization

The circulation is defined as:

$$\Gamma = \oint \mathbf{v} \cdot d\ell \tag{6}$$

We introduce the fundamental circulation quantum:

$$\Gamma_0 = 2\pi r_c \, \|\mathbf{v}_{\circlearrowleft}\| \tag{7}$$

and impose:

$$\Gamma = n\Gamma_0, \quad n \in \mathbb{Z} \tag{8}$$

[ORTHODOX]: Circulation quantization appears in superfluid systems.

[DERIVED]: In SST, Γ_0 is no longer an input but follows from Eq. 5.

2.4 Energy and Mass Relation

Mass is interpreted as stored rotational energy:

$$E = mc^2 \tag{9}$$

[ORTHODOX]: Standard relativistic energy relation.
[SPECULATIVE]: SST interprets this energy as localized vortex kinetic energy.

2.5 Swirl Velocity Constraint

We define the dimensionless ratio:

$$\eta_0 = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{c} \tag{10}$$

Empirically:

$$\eta_0 \approx \frac{\alpha}{2} \tag{11}$$

[DERIVED] Empirical closure: Within the present canonical calibration, this connects the characteristic swirl velocity to the fine-structure constant.

[CRITICAL NOTE] This relation should be read as a calibrated SST closure unless and until it is re-derived from a deeper dynamical sector.

2.6 Swirl-Clock Definition

We introduce the Swirl-Clock as a local scalar field encoding relativistic time dilation:

$$S_{(t)}^{\circlearrowleft} = \sqrt{1 - \frac{\|\mathbf{v}_{\circlearrowleft}\|^2}{c^2}} \tag{12}$$

[ORTHODOX] Functional form: This matches the standard Lorentz time-dilation factor.

[SPECULATIVE] Identification: In SST, the relevant velocity is interpreted as a local medium-kinematic state rather than only a standard worldline speed.

[DERIVED] Consequence: Once this identification is adopted, the Swirl-Clock provides a local scalar measure of kinematic time modulation within the medium description.

2.7 Density Hierarchy

We distinguish the following densities:

$$\rho_f \quad \text{(effective background fluid density)}, \tag{13}$$
$$\rho_E = \tfrac{1}{2}\rho_f \,\|\mathbf{v}_{\circlearrowleft}\|^2 \quad \text{(swirl energy density)}, \tag{14}$$
$$\rho_m = \rho_E/c^2 \quad \text{(mass-equivalent density)}, \tag{15}$$
$$\rho_{\text{core}} \quad \text{(saturated vortex-core density)}. \tag{16}$$

[ORTHODOX] Definition: $\rho_m = \rho_E/c^2$ is the mass-equivalent density associated with the swirl kinetic accounting above; in orthodox limits, total energy E in a region still yields mass density E/c^2.

[SPECULATIVE] Interpretation: The distinction between background, energetic, and saturated-core sectors is elevated in SST to a structural ontology of the medium.

2.8 Core Density Closure

The core density follows from a force constraint:

$$\rho_{\text{core}} \sim \frac{m_e c^2}{2\pi \|\mathbf{v}_{\circlearrowleft}\|^2 {r_c}^3} \tag{17}$$

[SPECULATIVE] Closure ansatz: Equation (17) is retained as the canonical core-density closure used by the present SST calibration program.

[DERIVED] Consequence: Within that closure, ρ_{core} is no longer an independent parameter.

2.9 Chronos–Kelvin Invariant and Clock–Radius Transport

For a thin material loop of radius $R(t)$ carried by an incompressible, inviscid flow with no reconnection events, Kelvin's theorem implies conservation of circulation. In the near-solid-body core approximation this gives

$$\frac{D}{Dt}(R^2\omega) = 0, \tag{18}$$

where ω is the characteristic local vorticity on the loop. Using the core relation $v_t = \omega r_c$ together with Eq. (12), one may rewrite

$$\omega = \frac{c}{r_c}\sqrt{1 - S_{(t)}^2}, \tag{19}$$

so that Eq. (18) takes the equivalent form

$$\frac{D}{Dt}\left(\frac{c}{r_c} R^2 \sqrt{1 - S_{(t)}^2}\right) = 0. \tag{20}$$

[ORTHODOX]: Eq. (18) is the thin-loop form of Kelvin circulation conservation in inviscid barotropic flow [6, 20].

[DERIVED]: Eq. (20) is the SST rewriting of the same invariant in terms of the Swirl-Clock.

Differentiating Eq. (20) gives a useful transport identity:

$$\frac{dS_{(t)}}{dt} = \frac{2\left(1 - S_{(t)}^2\right)}{S_{(t)}} \frac{1}{R}\frac{dR}{dt}. \tag{21}$$

[DERIVED]: contraction of a material loop drives stronger clock suppression, while expansion relaxes it.

2.10 Reparametrized Zero-Parameter Corollary

Older SST canon layers often used the triplet (Γ_0, ρ_f, r_c) as the primitive parameterization. In the present v0.8.0 architecture, Eq. (5) and Eq. (7) show that this is a reparameterization of the present primitive set $(\rho_f, \mathbf{v}_{\circlearrowleft}, \omega_c)$ rather than a distinct axiom.

Explicitly,

$$\Gamma_0 = 2\pi r_c \|\mathbf{v}_{\circlearrowleft}\| = 2\pi \frac{\|\mathbf{v}_{\circlearrowleft}\|^2}{\omega_c}. \tag{22}$$

Hence any dimensional SST quantity written in terms of (Γ_0, ρ_f, r_c) can be rewritten in terms of $(\rho_f, \mathbf{v}_{\circlearrowleft}, \omega_c)$ with no new input.

A coarse-grained density law retained from earlier canon layers is

$$\rho_f = K\,\Omega, \qquad K = \frac{\rho_{\rm core}\, r_c}{\|\mathbf{v}_{\circlearrowleft}\|}, \tag{23}$$

where Ω is a coarse-grained local rotation rate and K has dimensions $\mathrm{kg\,m^{-3}\,s}$. This relation records the canonical bridge between a sparse effective bulk density and a high-density localized core.

[DERIVED COROLLARY]: the older "zero-parameter" language is retained only as a statement of reparameterization discipline: once one primitive triplet is fixed, no additional dimensional constants should be introduced without explicit justification.

2.11 Canonical Master Equation

The preceding subsections define the four ingredients that later SST papers often treat separately: a local mass-equivalent density, a local Swirl-Clock, a geometric gate, and a topological kernel. We therefore introduce a single canonical equation that organizes these sectors in one place.

First define the local swirl energy density and its mass-equivalent form:

$$\rho_E(x) = \frac{1}{2}\rho_f(x)\,\|\mathbf{v}_{\circlearrowleft}(x)\|^2 \tag{24}$$

$$\rho_m(x) = \frac{\rho_E(x)}{c^2} \tag{25}$$

Using Eq. 12, the local Swirl-Clock factor is

$$S_{(t)}^{\circlearrowleft}(x) = \sqrt{1 - \frac{\|\mathbf{v}_{\circlearrowleft}(x)\|^2}{c^2}} \tag{26}$$

We then define a binary geometric exposure gate

$$G(K) \in \{0, 1\} \tag{27}$$

together with the canonical geometric factor

$$\Pi_K = \left(\frac{\lambda_c}{\pi r_c}\right)^{G(K)} = \left(\frac{4}{\alpha}\right)^{G(K)}. \tag{28}$$

The canonical topological kernel is taken to be

$$\Xi_K = [\alpha_C\, C(K) + \beta_L\, L(K) + \gamma_H\, \mathcal{H}(K)]\, \varphi^{-2k(K)}. \tag{29}$$

The canonical SST master equation is therefore

$$M_K(x) = \rho_m(x)\, {r_c}^3\, \Pi_K\, \Xi_K\, S_{(t)}^{\circlearrowleft}(x)^{-2} \tag{30}$$

$$= \left(\frac{\rho_f(x)\,\|\mathbf{v}_{\circlearrowleft}(x)\|^2}{2c^2}\right) {r_c}^3 \left(\frac{\lambda_c}{\pi r_c}\right)^{G(K)} [\alpha_C\, C(K) + \beta_L\, L(K) + \gamma_H\, \mathcal{H}(K)]\, \varphi^{-2k(K)} S_{(t)}^{\circlearrowleft}(x)^{-2} \tag{31}$$

[SPECULATIVE] Organizing principle: This equation factorizes SST into four sectors:

$$\text{medium scale} \times \text{geometric gate} \times \text{topological kernel} \times \text{clock impedance}. \tag{32}$$

[DERIVED] Dimensional check:

$$\left[\frac{\rho_f \, \|\mathbf{v}_{\circlearrowleft}\|^2}{c^2} {r_c}^3\right] = \mathrm{kg} \tag{33}$$

and all remaining factors are dimensionless, so Eq. 31 has the correct mass dimension.

[SPECULATIVE] Interpretation: The same local kinematic state that slows the Swirl-Clock also enters the effective mass scale; mass and local proper-time modulation are therefore treated as two aspects of one medium configuration.

[CRITICAL NOTE] Equation (31) is presently a canonical synthesis formula, not yet a theorem derived from first-principles filament dynamics alone.

In the stationary weak-swirl limit $S_{(t)}^{\circlearrowleft} \to 1$, it reduces to the static mass-functional form; in the local-kinematic limit at fixed K, it reduces to a pure clock-impedance correction. The hierarchy is therefore carried predominantly by the geometric and topological factors, while the Swirl-Clock supplies a coherent local correction rather than the primary source of mass separation.

2.12 Geometric Representation

A swirl-string is represented as a closed curve:

$$X(t,\sigma) = (x(t,\sigma), y(t,\sigma), z(t,\sigma)) \tag{34}$$

with $\sigma \in [0, 2\pi)$.

2.13 Summary of Dependency Structure

The full dependency chain is:

$$(\rho_f, \mathbf{v}_{\circlearrowleft}, \omega_c) \to r_c \to \Gamma_0 \to (\rho_E, \rho_m, \rho_{\mathrm{core}}) \to S_{(t)}^{\circlearrowleft} \to M_K. \tag{35}$$

[DERIVED] Consistency statement: No quantity is defined circularly. The canonical closure is summarized by Eq. 31.

2.14 Consistency Statement

- All derived quantities depend only on $\mathcal{P}$
- No external parameters are introduced
- The theory is closed under its primitive set

2.15 Interpretation

The Formal Foundations establish SST as:

- a continuum theory with minimal primitives
- a geometrically grounded framework
- a non-circular parameter system

3 Axiomatic Framework

3.1 Preliminaries

We formalize Swirl-String Theory (SST) as an axiomatic continuum theory defined over a three-dimensional spatial domain $\mathcal{M} \subset \mathbb{R}^3$ equipped with a preferred time parameter $t \in \mathbb{R}$.

The theory is constructed from a minimal primitive set $\mathcal{P}$ (Section 2) and a set of axioms $\mathcal{A}$ governing admissible configurations and dynamics.

3.2 Axiom 1: Continuum Substrate

Statement. There exists a continuous, incompressible, inviscid medium filling $\mathcal{M}$, characterized by a constant background density ρ_f.

$$\nabla \cdot \mathbf{v} = 0 \tag{36}$$

[ORTHODOX]: This corresponds to the standard incompressible Euler framework.

Interpretation. All physical structures arise as configurations within this medium; no discrete particles are assumed at the fundamental level.

3.3 Axiom 2: Existence of Stable Vortex Filaments

Statement. The medium admits stable, localized, topologically nontrivial vortex filament solutions.

[ORTHODOX] Physics input: vortex filaments are known idealized structures in fluid dynamics.

[SPECULATIVE] Extension: such filaments are assumed here to be dynamically stable and persist as fundamental carriers.

Mathematical form. A filament is represented as a closed embedding:

$$X : S^1 \to \mathcal{M}, \quad \sigma \mapsto X(t, \sigma) \tag{37}$$

3.4 Axiom 3: Circulation Quantization

Statement. Circulation around any stable filament is quantized:

$$\Gamma = \oint \mathbf{v} \cdot d\ell = n\Gamma_0, \quad n \in \mathbb{Z} \tag{38}$$

[ORTHODOX]: Quantized circulation appears in superfluid systems.

[SPECULATIVE]: Elevated here to a universal invariant.

3.5 Axiom 4: Compton Anchoring

Statement. The characteristic angular frequency of the vortex core satisfies:

$$\frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c} = \omega_c \tag{39}$$

[ORTHODOX]: ω_c is the Compton frequency.

[SPECULATIVE]: Identified as the intrinsic rotational frequency of the vortex core.

Consequence. The core radius is not independent but derived (Eq. 5).

3.6 Axiom 5: Delay as a Constitutive Principle

Statement. Finite propagation time along closed vortex structures is a fundamental dynamical ingredient.

$$\tau = \frac{L}{\|\mathbf{v}_{\circlearrowleft}\|} \tag{40}$$

[DERIVED] Kinematic consequence: the delay follows from finite signal speed along a closed carrier.

Axiomatic role. The delay is not treated as negligible, but as structurally relevant to the dynamics.

Interpretation. Temporal nonlocality is intrinsic to the dynamics.

3.7 Axiom 6: Energy Localization

Statement. Energy is localized in vortex structures and determines mass via:

$$E = mc^2 \tag{41}$$

[ORTHODOX]: Relativistic energy relation.
[SPECULATIVE]: Interpreted as rotational energy of the vortex configuration.

3.8 Axiom 7: Swirl-Clock Relational Time

Statement. Local time is encoded in a scalar field derived from fluid kinematics:

$$S_{(t)}^{\circlearrowleft} = \sqrt{1 - \frac{\|\mathbf{v}_{\circlearrowleft}\|^2}{c^2}} \tag{42}$$

[ORTHODOX] Functional form: this matches Lorentz time dilation.

[SPECULATIVE] Identification: the relevant velocity is taken to be the local medium kinematic state.

[DERIVED] Consequence: within SST this yields a relational clock field tied to local motion.

Interpretation. Time is not fundamental but relational to the state of motion in the medium.

3.9 Axiom 8: Density Saturation and Core Structure

Statement. The vortex core exhibits a saturated density ρ_{core} determined by mechanical constraints of the medium.

[SPECULATIVE] Closure hypothesis: the saturated density is fixed by the closure relation of Eq. (17).

[DERIVED] Consequence: once that closure is adopted, matter corresponds to regions where energy density reaches a structural limit.

3.10 Logical Structure of the Axioms

The axioms form a dependency hierarchy:

$$\text{Continuum} \rightarrow \text{Vortex Structures} \rightarrow \text{Quantization} \rightarrow \text{Delay Dynamics} \rightarrow \text{Mass/Energy Emergence} \tag{43}$$

3.11 Minimality Principle

Statement. The axioms are minimal in the sense that removing any one of them destroys at least one of the following:

- existence of stable structures
- emergence of discrete spectra
- compatibility with known physics

3.12 Philosophical Interpretation

The axiomatic structure implies:

- Particles are not fundamental objects but stable dynamical configurations
- Discreteness is emergent, not postulated
- Time is relational, not absolute

3.13 Consistency Condition

A valid SST realization must satisfy:

- Compatibility with orthodox physics in appropriate limits
- Internal logical closure
- Explicit traceability from axioms to predictions

3.14 Canonical Statement

Swirl-String Theory is defined as the minimal continuum theory in which stable vortex structures, governed by quantized circulation and delay dynamics, give rise to discrete physical phenomena consistent with known physics.

4 Consistency and Classification Framework

4.1 Classification Scheme

All statements are labeled as:

- **[ORTHODOX]** — established physics
- **[DERIVED]** — logically deduced within SST
- **[SPECULATIVE]** — novel or unverified

4.2 Extended Labeling Convention

We distinguish between *epistemic status* and *editorial role.*

Epistemic status. The only primary status labels used in the present Canon are:

- **[ORTHODOX]** — established in standard physics or mathematics
- **[DERIVED]** — obtained internally from the present SST assumptions, definitions, or closures
- **[SPECULATIVE]** — conjectural SST extension, interpretation, or unverified identification

Editorial role. Additional descriptors such as *Definition*, *Constraint*, *Interpretation*, *Benchmark*, *Roadmap*, and *Dimensional check* are not independent epistemic classes. They indicate only the role played by the statement.

Usage rule. When both are needed, the Canon uses the format

$$\texttt{[STATUS] Role: statement.} \tag{44}$$

For example, one may write **[ORTHODOX] Constraint**, **[DERIVED] Consequence**, or **[SPECULATIVE] Interpretation**.

4.3 Orthodox Constraints

The following must remain intact:

- Quantum mechanics predictions
- Atomic spectroscopy
- Relativistic consistency

4.4 Internal Consistency Rules

- No circular definitions
- All primitives explicitly defined
- Derived quantities must follow from axioms

4.5 Conflict Resolution Strategy

Priority order:

1. Logical consistency
2. Experimental agreement
3. Canonical simplicity

4.6 Epistemic Status Tracking

Each major result must explicitly state:

- Assumptions
- Derivation status
- Empirical testability

In particular, a definitional choice must not be presented as a theorem, and an interpretive bridge must not be presented as an orthodox result unless the underlying mathematical structure is itself standard.

4.7 Canon Integrity Principle

A valid SST canon must be internally consistent, externally compatible, and explicitly stratified by epistemic status.

5 Geometric and Topological Sector

5.1 Role of geometry and topology in the canon

This sector collects those parts of SST in which the identity of a state depends not only on local field amplitudes but on the global organization of the carrier. Geometry enters through lengths, radii, and closure constraints; topology enters through linking, crossing, chirality, and other invariant features that survive smooth deformations. At canon level, the main role of this sector is selective rather than exhaustive: it provides the organizing variables that feed the topological kernel Ξ_K and the geometric gate Π_K in Eq. (31).

The canon therefore treats geometry and topology as follows:

1. geometry sets admissible scales and shielding ratios,
2. topology labels persistent identity classes,
3. only those quantities that survive smooth deformation are allowed to enter the long-lived state classification.

This is also why layer or dressing indices are not treated as primary identity labels: they may modify a state, but they do not replace the topology-first classification of the carrier itself.

5.2 Spectral Structure of Locked Closed Carriers (Secondary Result)

Scope and status. This subsection records a *secondary consequence* of the kinematics of rationally locked, closed carriers. It does not introduce new primitives. All results are conditional on the existence of a well-defined phase field and a regular mode spectrum. Status: **derived under spectral assumptions; experimentally unverified**.

Kinematic reduction. Consider a closed carrier with a single central driving phase $\phi(t)$ and N_s sector variables $\theta_i(t)$ $(i = 1, \ldots, N_s)$ subject to a rational locking relation

$$\theta_i = \sigma_i \kappa_i \phi + \delta_i, \qquad \sigma_i \in \{+1, -1\}, \quad \kappa_i \in \mathbb{Q}. \tag{45}$$

In the rigid-lock limit, the dynamics reduce to a single collective coordinate $\phi(t)$ with effective Lagrangian

$$\mathcal{L}_{\text{eff}} = \frac{I_{\text{eff}}}{2}\dot{\phi}^2 - V(\phi), \tag{46}$$

where I_{eff} is an effective inertia and $V(\phi)$ is a periodic locking potential.

Distributed phase field. On a closed carrier of length L, allow small phase fluctuations

$$\phi(t) \;\longrightarrow\; \phi_0 + \eta(s,t), \qquad s \in [0, L], \tag{47}$$

with periodic boundary condition

$$\eta(s + L, t) = \eta(s, t). \tag{48}$$

A minimal quadratic field model consistent with Eq. (46) is

$$\mathcal{L}_{\text{field}} = \int_0^L \left[\frac{\mu_{\text{eff}}}{2}(\partial_t \eta)^2 - \frac{T_{\text{eff}}}{2}(\partial_s \eta)^2 - \frac{\mu_{\text{eff}}\omega_0^2}{2}\eta^2 \right] ds, \tag{49}$$

where μ_{eff} is an effective phase inertia density, T_{eff} an effective stiffness, and ω_0 the small-oscillation frequency set by $V(\phi)$.

Dimensional consistency:

$$[\mu_{\text{eff}}] = \text{kg m}, \qquad [T_{\text{eff}}] = \text{N}, \qquad [\omega_0] = \text{s}^{-1}.$$

Mode spectrum. Expanding in normal modes

$$\eta(s,t) = \sum_{n\in\mathbb{Z}} a_n(t)\, e^{ik_n s}, \qquad k_n = \frac{2\pi n}{L}, \tag{50}$$

yields the dispersion relation

$$\boxed{\omega_n^2 = \omega_0^2 + c_\phi^2 k_n^2, \qquad c_\phi^2 = \frac{T_{\text{eff}}}{\mu_{\text{eff}}}.} \tag{51}$$

For large n,

$$\omega_n \sim \frac{2\pi c_\phi}{L}\, n. \tag{52}$$

Cubic spectral sums and $\zeta(3)$. Consider a cubic-weighted spectral contribution of the form

$$\Delta\mathcal{E}^{(3)} = A_3 \sum_{n=1}^{\infty} \frac{1}{\omega_n^3}, \qquad [A_3] = \text{J s}^{-3}. \tag{53}$$

Using the asymptotic scaling (52),

$$\Delta\mathcal{E}^{(3)} \;\approx\; A_3 \left(\frac{L}{2\pi c_\phi}\right)^3 \sum_{n=1}^{\infty} \frac{1}{n^3}. \tag{54}$$

Hence

$$\boxed{\Delta\mathcal{E}^{(3)} \;\approx\; A_3 \left(\frac{L}{2\pi c_\phi}\right)^3 \zeta(3).} \tag{55}$$

Interpretation. Equation (55) shows that $\zeta(3)$ arises from the *high-mode spectral tail* of a closed carrier, not from the geometric threefold structure alone. The role of the locking relation (45) is indirect: it fixes I_{eff}, ω_0, and admissible boundary conditions, thereby determining the spectral scale (L, c_ϕ).

Universality. Different geometric realizations (e.g., distinct multi-sector carriers) that reduce to the same class of effective phase dynamics and share the asymptotic linear spectrum (52) will produce the same $\zeta(3)$ factor in Eq. (55), up to system-dependent prefactors.

$$\boxed{\zeta(3) \text{ is a spectral invariant of the mode tower, not a direct invariant of the carrier geometry.}} \tag{56}$$

Limitations and falsifiers. The result fails if:

1. the mode spectrum is not asymptotically linear in n,
2. strong damping or nonlocal coupling invalidates the expansion (49),
3. the system does not admit a well-defined periodic phase field on a closed domain.

Minimal experimental test. Measure mode frequencies ω_n for $n = 1, 2, 3, \ldots$ and verify

$$\frac{\omega_n}{n} \to \text{constant} \quad \text{as } n \to \infty. \tag{57}$$

Deviation from this scaling excludes Eq. (55).

Status. Secondary derived result. Valid under spectral assumptions; not a defining postulate.

5.3 Symmetry-Sensitive Dark-Knot Classification

A compact classification retained from earlier SST topology layers distinguishes visible, quasi-dark, and dark sectors by chirality, helicity, and low-order instability content.

Let $\chi(K)$ denote a chirality indicator, let

$$\mathcal{H}[K] = \int \mathbf{v} \cdot (\nabla \times \mathbf{v})\, dV \tag{58}$$

be the helicity functional, and let $N_u^{(1)}(K)$ count low-order unstable modes in a chosen perturbative stability analysis. Then the retained taxonomy is

$$\begin{aligned} \text{dark:} \quad & \chi(K) = 0, \quad \mathcal{H}[K] \approx 0, \quad N_u^{(1)}(K) = 0, && (59)\\ \text{quasi-dark:} \quad & |\chi(K)| \ll 1, \quad \mathcal{H}[K] \approx 0, \quad 0 < N_u^{(1)}(K) \leq 2, && (60)\\ \text{visible:} \quad & \chi(K) \neq 0 \quad \text{or} \quad \mathcal{H}[K] \neq 0. && (61) \end{aligned}$$

[ORTHODOX]: helicity is a standard invariant of ideal vortex dynamics [21, 6].

[SPECULATIVE]: the interpretation of Eq. (61) as a matter-sector taxonomy is an SST model choice.

6 Swirl–Electromagnetic Bridge

6.1 Minimal transverse field sector

A conservative bridge from the fluid sector to a radiation sector is obtained by introducing a divergence-free transverse field $\mathbf{A}_{\text{eff}}$ with

$$\nabla \cdot \mathbf{A}_{\text{eff}} = 0, \tag{62}$$

and effective Lagrangian density

$$\mathcal{L}_{\text{rad}} = \frac{\rho_f}{2} \|\partial_t \mathbf{A}_{\text{eff}}\|^2 - \frac{K_\gamma}{2} \|\nabla \times \mathbf{A}_{\text{eff}}\|^2. \tag{63}$$

Varying Eq. (63) and using Eq. (62) gives

$$\frac{1}{c_\gamma^2} \partial_t^2 \mathbf{A}_{\text{eff}} - \nabla^2 \mathbf{A}_{\text{eff}} = 0, \qquad c_\gamma^2 = \frac{K_\gamma}{\rho_f}. \tag{64}$$

Define

$$\mathbf{E}_{\text{eff}} = -\partial_t \mathbf{A}_{\text{eff}}, \qquad \mathbf{B}_{\text{eff}} = \nabla \times \mathbf{A}_{\text{eff}}. \tag{65}$$

Then one immediately recovers the source-free identities

$$\nabla \cdot \mathbf{B}_{\text{eff}} = 0, \tag{66}$$

$$\nabla \times \mathbf{E}_{\text{eff}} = -\partial_t \mathbf{B}_{\text{eff}}. \tag{67}$$

[ORTHODOX FORM]: Eq. (64) and Eq. (67) are the standard transverse-wave and source-free Faraday identities in vector-potential form [15].

[SPECULATIVE IDENTIFICATION]: SST identifies $\mathbf{A}_{\text{eff}}$ with a coarse-grained torsional or director displacement field of the medium.

6.2 Photon / unknot sector

The minimal retained radiation picture is that an R-phase excitation is an unknotted, finite-duration torsional packet propagating in the field $\mathbf{A}_{\text{eff}}$. At the Rosetta level this corresponds to a source-free transverse wave packet with dispersion inherited from Eq. (64).

[SPECULATIVE]: handedness of the local torsional packet is taken to encode polarization, while additional azimuthal phase winding encodes orbital angular momentum.

6.3 Swirl pressure law

For steady axisymmetric swirl, Euler radial balance gives

$$\frac{1}{\rho_f} \frac{dp_{\text{swirl}}}{dr} = \frac{v_\theta(r)^2}{r}. \tag{68}$$

For a locally rigid rotation profile $v_\theta(r) = \Omega r$, Eq. (68) integrates to

$$p_{\text{swirl}}(r) = p_0 + \frac{1}{2}\rho_f \Omega^2 r^2, \tag{69}$$

while for an exterior irrotational profile $v_\theta(r) = \Gamma/(2\pi r)$ one finds

$$p_{\text{swirl}}(r) = p_\infty - \frac{\rho_f \Gamma^2}{8\pi^2 r^2}. \tag{70}$$

[ORTHODOX]: Eq. (68) is the ordinary radial Euler balance for steady swirl [6, 20].

[DERIVED SST ROLE]: this pressure law is the canonical bridge between localized circulation and large-scale force analogies.

6.4 Minimal effective Lagrangian bridge

A compact field-theoretic summary retaining the clock, radiation, and matter sectors is

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_T + \mathcal{L}_{\text{rad}} + \sum_K \bar{\Psi}_K \big(i\gamma^\mu D_\mu - m_K^{(\text{sol})}\big)\Psi_K + \cdots, \tag{71}$$

where $\mathcal{L}_T$ is the clock/foliation sector introduced in Section 10, $\mathcal{L}_{\text{rad}}$ is given by Eq. (63), and the ellipsis denotes higher-order self-interaction, topological, or medium-response operators.

[SPECULATIVE BRIDGE]: Eq. (71) is a bookkeeping layer, not yet a completed first-principles derivation.

7 Delay and Mode Selection Theory

7.1 Circulation Delay as a Physical Primitive

We introduce the circulation delay as a constitutive element of the theory. For a closed vortex filament of effective length L and characteristic propagation speed $\|\mathbf{v}_{\circlearrowleft}\|$, the circulation time is defined as:

$$\tau = \frac{L}{\|\mathbf{v}_{\circlearrowleft}\|} \tag{72}$$

In the special case of a minimal closed loop associated with the vortex core scale, this reduces to:

$$\tau_c = \frac{2\pi r_c}{\|\mathbf{v}_{\circlearrowleft}\|} = \frac{2\pi}{\omega_c} \tag{73}$$

This establishes a direct identification between the circulation delay and the inverse Compton frequency.

[DERIVED]: The intrinsic delay of the vortex loop is identical to the Compton period.

7.2 Delayed Self-Interaction Dynamics

We model the phase $\phi(t)$ of a circulating excitation as a delayed self-interacting oscillator:

$$\dot{\phi}(t) = \omega_0 + \kappa \sin\big(\phi(t-\tau) - \phi(t)\big) \tag{74}$$

This equation represents a minimal effective description of a circulating field with finite propagation time.

[ORTHODOX]: Delay-differential equations of this type are well-established in feedback systems [1].

7.3 Emergence of Discrete Stable Modes

We seek steady-state solutions of the form:

$$\phi(t) = \Omega t \tag{75}$$

Substituting into Eq. 74 yields:

$$\Omega = \omega_0 + \kappa \sin(\Omega\tau) \tag{76}$$

This transcendental equation admits a discrete set of stable solutions Ω_n determined by the delay parameter τ.

[DERIVED]: Discrete mode selection arises from temporal nonlocality alone.

7.4 Delay-Induced Quantization Mechanism

Unlike conventional quantum mechanics, where discreteness is postulated, here it emerges dynamically:

- No boundary condition required
- No operator quantization assumed
- Discreteness arises from stability selection

[SPECULATIVE]: Particle spectra may be interpreted as delay-selected stable circulation modes.

7.5 Extension to Nonlinear Schrödinger Dynamics

To connect with field-theoretic descriptions, we extend the delay mechanism to a nonlinear Schrödinger framework:

$$i\hbar\partial_t\psi = -\frac{\hbar^2}{2m}\nabla^2\psi + F_{\mathrm{delay}}[\psi(t-\tau)] \tag{77}$$

where the delayed functional introduces temporal nonlocality into the field evolution.

[SPECULATIVE]: This provides a bridge between classical delay systems and quantum wave dynamics.

7.6 Physical Interpretation

The delay τ acts as:

- a memory kernel
- a nonlocal temporal coupling
- a selection mechanism for stable modes

Thus:

$$\textit{Delay} \Rightarrow \textit{Stability} \Rightarrow \textit{Discreteness}$$

This mechanism forms one of the central pillars of the SST canon.

8 Atomic Bridge Model

8.1 Introduction and epistemic framing

This section records the minimal SST atomic bridge rather than a full derivation of atomic structure. Its canon role is deliberately narrow: recover the orthodox Coulomb limit when the SST-specific sector is switched off, define a branch-resolved circulation coupling fixed by canonical medium scales, and make explicit which many-electron and shell-structure claims remain open. It is therefore a benchmark layer, not yet a complete atomic theory.

8.2 Baseline Hamiltonian

We adopt the standard Coulomb Hamiltonian as the orthodox baseline:

$$H_0 = -\frac{\hbar^2}{2\mu}\nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 r} \tag{78}$$

[ORTHODOX]: This reproduces the hydrogenic spectrum.

8.3 Legacy Soft-Core Regularization

Early canon layers introduced a finite-core regularization of the hydrogenic potential. In the present document this is retained only as a controlled bridge ansatz, not as a replacement for Eq. (78).

Define the effective coupling scale

$$\Lambda_{\text{SST}} = 4\pi\,\rho_{\text{core}}\,\|\mathbf{v}_{\circlearrowleft}\|^2\,{r_c}^4 \tag{79}$$

with dimensions

$$[\Lambda_{\text{SST}}] = \text{J}\,\text{m} = \text{N}\,\text{m}^2. \tag{80}$$

A finite-core bridge potential is then

$$V_{\text{soft}}(r) = -\frac{\Lambda_{\text{SST}}}{\sqrt{r^2 + r_c^2}} \tag{81}$$

satisfying the large-radius limit

$$V_{\text{soft}}(r) \xrightarrow{r \gg r_c} -\frac{\Lambda_{\text{SST}}}{r}. \tag{82}$$

[ORTHODOX] Effective-model input: finite-core regularizations of singular central potentials are standard tools.
[DERIVED] Canonical consequence: the SST scale in Eq. (79) is fixed by canonical constants rather than introduced phenomenologically.
[ORTHODOX] Constraint: this sector is admissible only if the large-radius limit reproduces the standard Coulomb problem to spectroscopic accuracy.

In particular, Eq. (81) is retained as a bridge ansatz only. It is not promoted here to the primary atomic potential.

8.4 SST Circulation Correction

We introduce a circulation-dependent perturbation:

$$\delta H^{(\sigma)} = -2\sigma E_R \eta_0 \gamma + \frac{E_R \eta_0 \sigma}{2}\left\{\frac{\hat{L}_z}{\hbar}, \gamma\right\} \tag{83}$$

[DERIVED]: Interaction depends on circulation amplitude $\gamma(r,t)$.

8.5 Circulation source chain

The atomic perturbation is not introduced as an arbitrary extra scalar, but is intended to descend from a circulation-bearing source chain,

$$\xi \to \omega_z \to u_\theta \to \Gamma_{\text{loc}} \to \gamma, \tag{84}$$

where an axial displacement field ξ generates local vorticity, the vorticity sources azimuthal swirl, and the loop circulation is normalized by the canonical quantum Γ_0. This makes the circulation amplitude γ the natural canon-level coupling variable rather than a free phenomenological insertion.

8.6 Branch Structure

Two chiral sectors exist:

$$\sigma = \pm 1 \tag{85}$$

These correspond to:

- ↺ (left-handed)
- ↻ (right-handed)

8.7 Consistency Requirement

To remain compatible with atomic physics:

$$|\delta E_n| \ll |E_n| \tag{86}$$

[ORTHODOX] Constraint: SST corrections must not violate known spectroscopy.

8.8 Interpretation

[STATUS]: The atomic bridge is a controlled benchmark sector. The orthodox Coulomb problem is imported as the baseline; the circulation-dependent term is the SST addition; shell counting, exclusion-like behavior, and heavy-atom extensions remain programmatic rather than derived here. The purpose of the section is therefore structural compatibility, not yet a complete replacement of atomic quantum mechanics.

9 Spectroscopic Constraints

9.1 Motivation

Any extension of atomic structure must remain consistent with high-precision spectroscopic measurements.

[ORTHODOX]: Atomic energy levels are experimentally verified to extremely high precision.

Requirement:

$$|\delta E_n| \ll |E_n| \tag{87}$$

where δE_n represents any SST-induced correction.

9.2 Internal Mode Hypothesis

We consider the possibility that atomic bound states possess an additional internal excitation sector associated with vortex filament dynamics.

[SPECULATIVE]: Each bound state carries an internal mode spectrum $\{E_{m,n}\}$.

The effective energy becomes:

$$E_n^{\text{eff}} = E_n^{(0)} + \delta E_n \tag{88}$$

9.3 Generic Coupling Structure

We parameterize the coupling between internal modes and observable energy levels:

$$|\delta E_n| \leq \lambda_n \sum_m E_{m,n}\, \nu_{m,n} \tag{89}$$

where:

- λ_n is the coupling efficiency
- $\nu_{m,n}$ is the occupation factor

[DERIVED]: This form captures a general upper bound independent of microscopic details.

9.4 Ungapped Spectrum: Instability

If the internal spectrum is ungapped:

- arbitrarily low-energy excitations exist
- no universal suppression mechanism applies

Consequence:

$$\delta E_n \not\to 0 \quad \text{generically} \tag{90}$$

[CRITICAL]: An ungapped internal sector is generically incompatible with spectroscopy unless:

- $\lambda_n \ll 1$ (extremely weak coupling), or
- the density of states is highly suppressed

9.5 Gapped Spectrum: Decoupling Mechanism

We introduce an excitation gap Δ_K:

$$E_{m,n} \geq \Delta_K \tag{91}$$

Then occupation factors follow Boltzmann suppression:

$$\nu_{m,n} \sim \exp\left(-\frac{\Delta_K}{k_B T_{\text{eff}}}\right) \tag{92}$$

Substituting into Eq. 89:

$$\delta E_n \propto \exp\left(-\frac{\Delta_K}{k_B T_{\text{eff}}}\right) \tag{93}$$

[DERIVED]: Exponential suppression ensures compatibility with spectroscopy.

9.6 Physical Origin of the Gap

[SPECULATIVE]: The gap Δ_K may arise from:

- geometric constraints on filament curvature
- torsional rigidity
- topological restrictions
- core structure of the vortex

9.7 Order-of-Magnitude Estimate

Using filament parameters:

$$\Delta_K \sim \frac{\Gamma^2}{4\pi\xi L} \tag{94}$$

where:

- ξ is the core scale
- L is the loop length

For typical SST scales:

$$\Delta_K = \mathcal{O}(10^2 - 10^3\,\mathrm{eV}) \tag{95}$$

[DERIVED]: This places the internal sector well above atomic energy scales ($\sim$ eV).

9.8 Consistency Condition

To preserve atomic physics:

$$\frac{\Delta_K}{k_B T_{\mathrm{eff}}} \gg 1 \tag{96}$$

Interpretation: The internal sector is effectively frozen at atomic energies.

9.9 Observable Consequences

[SPECULATIVE]: Residual effects may include:

- tiny level shifts
- suppressed line broadening
- weak temperature dependence

9.10 Falsifiability

The theory is falsifiable via:

- detection of unexplained spectral shifts
- absence of expected suppression
- deviation from Boltzmann scaling

9.11 Integration with SST Framework

The spectroscopic constraint enforces:

- compatibility with the Atomic Bridge (Section 8)
- constraints on delay-induced excitations (Section 7)
- bounds on allowable vortex configurations

9.12 Canonical Statement

Any viable SST extension must include a mechanism—such as an excitation gap—that ensures exponential suppression of internal modes at atomic energy scales.

9.13 Precision electroweak bridge: lessons from the CMS W-mass measurement

Status. [ORTHODOX] External input: the CMS measurement is a precision constraint on any SST effective weak-sector realization. **[DERIVED]** Methodological consequence: the result supports a forward-modelling strategy for SST phenomenology. **[CRITICAL NOTE]** The paper does *not* provide direct evidence for a structured-substrate ontology of the W boson.

Empirical anchor. Let the experimentally inferred resonance mass be denoted by

$$m_W^{\rm exp} = 80.3602 \pm 0.0099 \text{ GeV}. \tag{97}$$

In the CMS analysis this quantity is not read off from a single observable, but extracted from a profiled likelihood over finely binned charged-lepton kinematics. This motivates the following SST principle:

> *Collider observables constrain SST through forward maps from latent swirl-string states to detector-level distributions, not through direct identification of a single internal parameter with a measured mass.*

Latent-to-observable map. Introduce a latent SST parameter set for an effective spin-1 weak excitation,

$$\Theta_W^{\rm SST} = \Big(K_W,\, \chi_W,\, \mathcal{L}_W,\, \mathcal{H}_W,\, S_{(t)W}^{\circlearrowleft},\, \rho_f^{\rm eff},\, \mathbf{v}_{\circlearrowleft}^{\rm eff}\Big), \tag{98}$$

where:

- K_W is the topology class or excitation family,
- χ_W is chirality / handedness label,
- $\mathcal{L}_W$ is an effective geometric length scale,
- $\mathcal{H}_W$ is a helicity/linkage content,
- $S_{(t)W}^{\circlearrowleft}$ is the local clock factor for the mode,
- $\rho_f^{\rm eff}$ and $\mathbf{v}_{\circlearrowleft}^{\rm eff}$ are the coarse-grained medium parameters seen by the excitation.

The experimentally accessible distribution is then not a bare function of m_W alone, but a projected density

$$\mathcal{P}_{\rm meas}(x) = \int dz\; R(x|z)\, \mathcal{P}_{\rm prod}(z \mid \Theta_W^{\rm SST}), \tag{99}$$

where z denotes truth-level production/decay variables and $R(x|z)$ is the detector response kernel.

Mass as an effective resonance parameter. In SST one should distinguish:

$$m_W^{\text{internal}} \neq m_W^{\text{eff}} \tag{100}$$

in general, where:

- m_W^{internal} is the internal excitation-energy scale of the structured mode,
- m_W^{eff} is the resonance parameter inferred from scattering distributions.

The experimentally constrained quantity is m_W^{eff}.

A minimal canon-safe matching condition is therefore

$$m_W^{\text{eff}}(\Theta_W^{\text{SST}}) = m_W^{\text{exp}} + \delta_W, \qquad |\delta_W| \lesssim 10 \text{ MeV}, \tag{101}$$

with the current precision scale set by experiment.

Distribution-level fit target. The correct SST comparison object is a likelihood built from binned observables:

$$\mathcal{L}\big(\Theta_W^{\text{SST}}, \nu\big) = \prod_{b=1}^{N_{\text{bins}}} \text{Poisson}\big(n_b \,\big|\, \mu_b(\Theta_W^{\text{SST}}, \nu)\big)\ \Pi(\nu), \tag{102}$$

where:

- n_b is the observed count in bin b,
- μ_b is the SST prediction after production, decay, and detector folding,
- ν denotes nuisance parameters,
- $\Pi(\nu)$ is the nuisance prior / constraint factor.

The profile likelihood for the physically interesting SST parameters is then

$$\mathcal{L}_{\text{prof}}(\Theta_W^{\text{SST}}) = \max_{\nu}\ \mathcal{L}(\Theta_W^{\text{SST}}, \nu). \tag{103}$$

Helicity decomposition as a falsifier. A particularly sharp SST test is not the central value of m_W alone, but whether the effective spin-1 mode reproduces the standard angular basis. Write

$$\frac{d\sigma}{d\Omega} = \sum_{i=0}^{8} A_i\, f_i(\theta, \phi) + \epsilon_{\text{SST}}\, \Delta(\theta, \phi), \tag{104}$$

where the $A_i f_i$ are the standard helicity structures and $\epsilon_{\text{SST}}\Delta$ is the first SST correction. Then:

- if $\epsilon_{\text{SST}} = 0$ within experimental precision, SST survives this channel,
- if $\epsilon_{\text{SST}} \neq 0$, the deviation is directly falsifiable through angular distributions.

Canon interpretation. This leads to the following conservative reading:

1. The weak sector must be *SM-like at the effective distribution level.*
2. Large $\mathcal{O}(10^{-4})$–$\mathcal{O}(10^{-3})$ relative distortions in the effective W mass are disfavoured.
3. The best SST discovery channels are therefore likely to be

$$\text{small angular deviations}, \quad \text{polarization asymmetries}, \quad \text{off-shell tails}, \quad \text{rare differential distortions}, \tag{105}$$

   rather than a large shift in the central value of m_W.

Minimal bridge claim. The CMS result does *not* force the W boson to be fundamental in the ontological sense. It does force any SST realization of the weak sector to satisfy the stronger requirement:

$$\mathcal{P}_{\mathrm{prod}}^{\mathrm{SST}}(z \mid \Theta_W^{\mathrm{SST}}) \approx \mathcal{P}_{\mathrm{prod}}^{\mathrm{SM}}(z \mid m_W, \mathrm{SM\ nuisances}) \tag{106}$$

to current experimental precision after detector folding and nuisance profiling.

Practical SST program. A precision-compatible SST weak-sector program should therefore proceed in three layers:

$$\text{(i) effective resonance matching:} \quad m_W^{\mathrm{eff}} \to m_W^{\mathrm{exp}}, \tag{107}$$

$$\text{(ii) helicity matching:} \quad A_i^{\mathrm{SST}} \to A_i^{\mathrm{SM}}, \tag{108}$$

$$\text{(iii) residual search:} \quad \Delta \neq 0 \text{ only in subleading differential structure.} \tag{109}$$

Conclusion. The CMS W-mass result is best understood in SST as a *precision filter*: it does not supply ontology, but it strongly constrains phenomenology. Any viable SST weak-sector realization must be distributionally SM-like to present precision, while leaving room only for small, structured residuals in higher-order observables.

10 Relational Time Framework

10.1 Swirl-Clock and local proper time

The canonical time variable of SST is not introduced as an independent primitive. It is defined relationally from the local kinematic state of the medium through the Swirl-Clock already introduced in Eq. (12). In its coarse-grained form,

$$dt_{\mathrm{loc}} = S_{(t)}(x)\, dt_{\infty}, \qquad S_{(t)}(x) = \sqrt{1 - \frac{v(x)^2}{c^2}}. \tag{110}$$

Here t_∞ denotes the asymptotic laboratory time and t_{loc} the effective proper time associated with a localized process in the medium. In SST, this factor is interpreted as a fluid-kinematic clock field rather than as a purely geometric spacetime postulate.

10.2 Carrier-local clocking and the turnover scale

For localized carriers the natural microscopic rate is the turnover frequency

$$\Omega_0 = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c} = \omega_c, \tag{111}$$

which coincides with the Compton anchoring scale. A complete SST clock description therefore involves two linked but distinct levels:

- the *microscopic turnover scale* Ω_0, which counts circulation cycles of the carrier,
- the *coarse-grained Swirl-Clock* $S_{(t)}$, which relates local rates to asymptotic time.

This resolves a recurring ambiguity in earlier canon versions: local cycle counting and coarse-grained time dilation are related, but they are not identical observables.

10.3 Relational observables from conserved currents

A purely operator-based notion of time is avoided in SST by defining time observables through conserved currents. Let J^μ denote a matter current and let $j^\mu_{\rm ev}$ denote a coarse-grained event current associated with structured activity of the medium. The relational clock field $T(x)$ is then recovered only after coarse graining,

$$\partial_\mu T(x) \approx \frac{1}{\nu_0}\,\langle j^{\rm ev}_\mu\rangle_\ell, \tag{112}$$

where ℓ is the coarse-graining scale and ν_0 a reference event frequency. The canon interpretation is that time is read from correlations between currents, not from an abstract external operator.

10.4 Time-of-arrival as a relational field observable

Let $\mathcal{W}$ be a detector world-tube with boundary $\Sigma = \partial\mathcal{W}$. Then the arrival-time distribution associated with a detector label Θ is written as

$$p(\Theta) = \frac{1}{\mathcal{N}} \int_\Sigma d\sigma_\mu\, J^\mu(x)\,\delta\big(T(x) - \Theta\big). \tag{113}$$

This formulation is canonically important because it bypasses the need for a self-adjoint time operator conjugate to a bounded Hamiltonian. The measured time is instead the value of a physical clock field at the point where matter flux crosses the detector boundary.

10.5 Clock broadening and continuum limits

If the event current has a finite variance, then the observed time-of-arrival distribution is broadened relative to the ideal classical prediction. Writing $P_{\rm cl}$ for the classical arrival distribution and σ_τ^2 for the effective clock variance, one expects a convolution structure of the form

$$P_{\rm obs}(\Theta) = \int dt\, P_{\rm cl}(t) \frac{1}{\sqrt{2\pi\sigma_\tau^2}} \exp\left[-\frac{(\Theta - t)^2}{2\sigma_\tau^2}\right]. \tag{114}$$

In the limit $\ell \gg r_c$, the event structure is smoothed and one recovers a continuum clock field. In the opposite limit $\ell \to r_c$, the underlying discreteness of carrier events becomes operationally relevant.

10.6 Connection to gravity and geometry

In the canon, the relational-time sector and the geometric-response sector are not independent. Variations in the Swirl-Clock track variations in local swirl energy density and therefore provide the time component of the effective geometric response of the medium. At this level, the SST reading is:

gravity and time are two coarse-grained aspects of the same clock-bearing medium.

This statement should be read as a canon-level organizing principle, not yet as a completed field-theoretic proof.

10.7 Clock Field and Foliation Variable

[DERIVED] EFT bridge: the scalar Swirl-Clock field is promoted from the kinematic relation of Eq. (12) to a long-wavelength foliation variable

$$\chi(x), \tag{115}$$

whose normalized gradient defines a preferred local clock frame:

$$u_\mu = \frac{\nabla_\mu \chi}{\sqrt{g^{\alpha\beta}\nabla_\alpha \chi \, \nabla_\beta \chi}}. \tag{116}$$

[ORTHODOX] Equation (116) matches the standard hypersurface-orthogonal foliation construction used in khronometric / Einstein–Æther effective field theory [7, 8].

[DERIVED] Within SST, the interpretation is different: u_μ is not taken as a fundamental spacetime structure, but as the coarse-grained clock-flow direction associated with the medium.

10.8 Clock-Sector Effective Action

[ORTHODOX] EFT template: the most general second-order covariant clock-sector action compatible with hypersurface orthogonality is

$$\begin{aligned} S_\chi = \int d^4x\sqrt{-g}\,\Big[&c_1(\nabla_\mu u_\nu)(\nabla^\mu u^\nu) + c_2(\nabla_\mu u^\mu)^2 \\ &+ c_3(\nabla_\mu u_\nu)(\nabla^\nu u^\mu) + c_4 u^\mu u^\nu(\nabla_\mu u_\alpha)(\nabla_\nu u^\alpha)\Big]. \end{aligned} \tag{117}$$

[SPECULATIVE] SST interpretation: Equation (117) is not itself derived from first-principles filament dynamics in the present Canon. It is recorded as the minimal orthodox EFT bridge into a relativistic clock-field description.

10.9 Poisson Limit and Effective Gravity Bridge

[DERIVED] Bridge statement: in the weak-field, slowly varying limit, the scalar clock field is taken to obey a Poisson-type closure

$$\nabla^2\chi(x) = 4\pi G_{\text{eff}}\,\rho_{\text{matter}}(x), \tag{118}$$

so that outside localized sources

$$\chi(r) \propto \frac{1}{r}, \qquad \mathbf{g}_{\text{eff}} = -\nabla\chi. \tag{119}$$

[ORTHODOX] The $1/r$ scalar potential and $1/r^2$ force law are standard consequences of Eq. (118).

[SPECULATIVE] The identification of χ with the Swirl-Clock and the interpretation of Eq. (118) as an emergent gravity law are SST-specific.

10.10 Observational Constraint from Tensor-Mode Speed

[ORTHODOX] Constraint: If the clock-sector EFT is used, multimessenger gravitational-wave observations require [9]

$$c_{13} \equiv c_1 + c_3 = 0, \tag{120}$$

to enforce luminal tensor-mode propagation at observational precision.

[ORTHODOX] Constraint: any SST realization employing the EFT bridge of Eq. (117) must preserve Eq. (120).

10.11 Interpretive Role in the Canon

This section has a deliberately limited role:

- it upgrades the Swirl-Clock from a purely local kinematic factor to a candidate field variable;
- it records a minimal orthodox EFT bridge for relativistic comparison;
- it supplies the internal reference point needed by Section 11 for the time–gravity link.

Status summary.

- Eq. (116): **[ORTHODOX]** mathematical form; **[DERIVED]** role as an SST bridge variable
- Eq. (117): **[ORTHODOX]** EFT template
- Eq. (118): **[DERIVED]** bridge closure
- gravity interpretation: **[SPECULATIVE]**

10.12 Relation to the Chronos–Kelvin law

Section 2.9 provides the local transport identity of the Swirl-Clock on a material loop. The present section supplies the long-wavelength field-theoretic and analogue-metric reading of the same clock sector. The two layers are therefore complementary:

- Eq. (21) is the loop-level transport law;
- Eq. (116)–Eq. (118) provide the coarse-grained field description.

11 Unified Interpretation

11.1 Overview

Swirl-String Theory (SST) proposes a unified interpretation in which fundamental physical phenomena emerge from the dynamics of a continuous medium supporting stable vortex structures.

This section synthesizes the results of the previous sections into a coherent conceptual framework.

11.2 Ontology of the Theory

[SPECULATIVE] Ontological claim: The fundamental ontology consists of:

- A continuous incompressible medium (Section 3)
- Stable vortex filament configurations (Section 5)
- Circulation as a conserved topological invariant

Interpretation: Particles are not elementary objects, but persistent dynamical structures.

11.3 Mass as Trapped Circulation Energy

From Section 2, and in particular from Eq. (31), the mass sector is organized as the product of a medium scale, a geometric gate, a topological kernel, and a local clock-impedance factor. The canon-level reading is therefore not merely that "energy is stored in rotation," but that persistent inertial scale is assigned only after geometry, topology, and local clocking have all been specified.

$$M_K(x) = \rho_m(x)\, {r_c}^3\, \Pi_K\, \Xi_K\, S^{\circlearrowleft}_{(t)}(x)^{-2}. \tag{121}$$

[SPECULATIVE] Unified interpretation: mass corresponds to localized, self-sustained circulation energy in the medium, modulated by state geometry, topology, and local clock impedance.

11.4 Time as a Relational Quantity

From Section 10 and Section 2, time is not introduced as an independent primitive but as a relational field read from the local state of the medium. The same kinematic quantity that enters the Swirl-Clock also appears in the organizing equation of the mass sector, so the canon treats proper-time modulation and inertial scaling as two projections of one shared substrate state.

$$S^{\circlearrowleft}_{(t)}(x) = \sqrt{1 - \frac{\|\mathbf{v}_{\circlearrowleft}(x)\|^2}{c^2}}. \tag{122}$$

[SPECULATIVE] Unified interpretation: time is not fundamental, but emerges from the kinematic state of the medium and is read operationally through clock-bearing processes.

11.5 Discreteness without Quantization Postulates

From Section 7:

- Delay introduces temporal nonlocality
- Stability selects discrete modes

[DERIVED] Unified consequence:

Discrete physical spectra arise dynamically from delay-induced mode selection.

This replaces operator-based quantization with a dynamical mechanism.

11.6 Charge and Circulation

[SPECULATIVE] Provisional bridge only. The canon retains circulation as a structurally important invariant, but it does *not* yet claim a completed derivation of observed gauge charges from circulation data alone. At most, the present document records a working hypothesis that orientation and magnitude of circulation may enter a later charge-assignment program.

11.7 Unification of Interactions

The canon records a *unification program*, not yet a completed unification theorem. In the present document:

- electromagnetic effects are represented by the minimal transverse-field bridge of Section 6,
- gravitational effects are represented by the clock-field and Poisson-limit bridge of Section 10,
- inertial mass is organized by the medium/topology/geometry/clock structure of Eq. (31).

[SPECULATIVE] Programmatic claim: these sectors may ultimately be read as different effective descriptions of one substrate, but their full common derivation remains open.

11.8 Atomic Structure as an Emergent Layer

From Section 8:

- Standard quantum mechanics appears as an effective description
- SST introduces circulation-dependent corrections

[SPECULATIVE] Unified interpretation:

Quantum mechanics is an emergent effective theory of underlying vortex dynamics.

11.9 Consistency with Spectroscopy

From Section 9:

- Internal modes are suppressed by an excitation gap
- Low-energy physics remains unchanged

[ORTHODOX] Constraint: The unified picture does not violate known experimental constraints.

11.10 Hierarchy of Description

The theory admits multiple levels:

1. **Microscopic level:** vortex filament dynamics
2. **Mesoscopic level:** delay-induced mode structure
3. **Macroscopic level:** effective quantum description

11.11 Conceptual Summary

The SST framework can be summarized as:

Mass	$\rightarrow$ trapped circulation energy
Time	$\rightarrow$ relational swirl-clock field
Charge	$\rightarrow$ provisional circulation bridge
Quantization	$\rightarrow$ delay-induced stability

11.12 Canonical Statement

All observed physical structure emerges from the dynamics of a continuous medium in which stable vortex configurations, governed by quantized circulation and delay dynamics, produce the phenomena conventionally attributed to particles, fields, and quantum mechanics.

11.13 Scope and Limits

[IMPORTANT]:

- The framework is not yet a complete fundamental theory
- Several mechanisms remain to be derived rigorously
- Many results are conditional or phenomenological

11.14 Final Consistency Principle

A valid unified interpretation must preserve all experimentally verified results while providing a deeper structural explanation of their origin.

12 Integration of Prior Canon Versions

12.1 Integration policy (v0.5.12 reintegration)

This block records only material from earlier SST canon layers that can be retained inside the v0.8.0 architecture without weakening the explicit classification discipline of Section 4.

Older canon documents are therefore *not* cited as authorities. Transferable content is rewritten here as current-canon statements, each labeled according to its present status.

12.2 Retained v0.5.12 kernel sectors

The present draft now retains the following v0.5.12 kernel sectors in rewritten form:

- the Chronos–Kelvin invariant and clock–radius transport law (Section 2.9);
- the reparametrized zero-parameter corollary and coarse-graining rule (Section 2.10);
- a minimal Swirl–EM bridge and photon / unknot sector (Section 6);
- the swirl pressure law as the Euler-level force bridge (Section 6.3);
- the analogue-metric / clock-field bridge (Section 10).

12.3 Wave–particle phase rule and measurement bridge

A retained qualitative rule from earlier canon layers is that SST distinguishes two limiting phases:

$$\text{R-phase}: \quad \text{extended, unknotted, radiative,} \tag{123}$$

$$\text{T-phase}: \quad \text{localized, knotted, tangible.} \tag{124}$$

A minimal dynamical bookkeeping law for phase conversion is

$$\partial_t P_T = \Gamma_{R\to T}[\mathcal{K}]\, P_R - \Gamma_{T\to R}[\mathcal{K}]\, P_T, \qquad P_R + P_T = 1, \tag{125}$$

where $\mathcal{K}$ denotes an environment- and coherence-dependent kernel functional. The intended interpretation is that "measurement" corresponds not to an axiomatically imposed collapse, but to a medium-induced transfer of support from the extended R-phase to the localized T-phase.

[ORTHODOX LIMIT]: in weak coupling and after coarse graining, Eq. (125) should reduce to ordinary environment-induced decoherence models [22].

[SPECULATIVE]: the identification of decoherence with an actual R↔T phase transition is SST-specific.

12.4 Gauge and weak-mixing bridge

The retained gauge-level statement is that the effective low-energy algebra should reduce to

$$\mathfrak{g}_{\text{eff}} = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1), \tag{126}$$

with gauge potentials interpreted as holonomy data of coarse-grained multi-director transport.

A minimal bridge ansatz for the weak mixing angle is

$$\sin^2 \theta_W^{\text{eff}} = \frac{K_Y}{K_Y + K_W}, \tag{127}$$

where K_Y and K_W are effective Abelian and weak-sector stiffness scales. Eq. (127) is not a completed derivation, but it preserves the v0.5.12 idea that weak mixing should descend from a ratio of medium-response coefficients rather than appear as an unstructured free number.

[ORTHODOX]: θ_W is the standard electroweak mixing parameter of the SM [23, 3].

[SPECULATIVE]: Eq. (127) is an SST bridge ansatz only.

12.5 What the v0.5.12 reintegration does not import

The present reintegration does *not* import v0.5.12 cosmogony, engineering subsystems, cosmological-term closures, or highly application-specific protocols into the main canon body. Those remain suitable for appendices or standalone papers until they are more tightly tied to the minimal axiom system.

12.6 Purpose of this integration layer

Version 0.8.0 is not a fresh theory written from zero, but an explicit consolidation of earlier canon lines. This section records what has been retained, what has been renamed or reorganized, and what has been demoted from canon status to research-track status.

12.7 Retained from the early canon layers

From the v0.1.x–v0.3.x line, the present canon retains the primitive closure logic:

- a continuum substrate with effective density ρ_f,
- a core scale r_c fixed by Compton anchoring,
- circulation quantization through $\Gamma_0 = 2\pi r_c \,\|\mathbf{v}_{\circlearrowleft}\|$,
- the interpretation of matter as stable organized structures rather than point primitives.

These elements remain non-negotiable because later sectors depend on them directly.

12.8 Retained from the middle canon layers

From the v0.4.x–v0.6.x line, the canon retains three durable structures:

1. the Rosetta strategy for mapping SST statements into orthodox language,
2. the photon/torsion sector as the minimal radiation bridge,
3. delay-induced mode selection as the main route to discreteness without operator postulates.

These are now distributed across Sections 7, 10, and 11 rather than treated as isolated notes.

12.9 Retained from the late canon layers

From the v0.7.x line, v0.8.0 retains the strongest structurally reusable sectors:

- the Swirl-Clock as the canonical relational-time field,
- the atomic bridge as a benchmark consistency layer rather than a complete atomic derivation,
- the spectroscopic gap logic as a hard consistency filter,
- the topology-first mass program, with topology separated from thermodynamic or radiative dressing,
- the EFT-style foliation and gauge-emergence program as a long-range development track.

12.10 Notation and density cleanup

Earlier canon versions sometimes conflated background density, mass-equivalent density, and saturated core density. The present version adopts the cleaned hierarchy

$$\rho_f \quad \text{(effective fluid density)}, \qquad \rho_E \quad \text{(swirl energy density)}, \qquad \rho_m = \rho_E/c^2, \qquad \rho_{\text{core}} \quad \text{(saturated core densit} \tag{128}$$

Any older passage that identifies a single symbol with more than one of these roles is superseded by the present separation.

12.11 Topology-first reinterpretation of layers

A major consolidation from the recent lepton and knot-mass program is the separation between topology and layering. Knot type is retained as the primary identity label, while Golden-layer or thermal ladder indices are treated as dressing or coherence variables rather than species-defining labels. This prevents the canon from over-reading numerical layer fits as ontological identities.

12.12 Status of optional sectors

The following sectors are retained as important but non-primitive research layers:

- full gauge emergence and charge assignments,
- dual-vacuum phenomenology and Unruh-echo interpretation,
- complete many-electron atomic closure,
- full weak-sector and collider phenomenology,
- explicit topology-to-particle dictionary beyond the best-supported benchmark assignments.

They remain in the canon as organized programs, not as closed theorem-level results.

12.13 Integration principle

The purpose of v0.8.0 is therefore not to maximize novelty per page, but to minimize duplication and to expose the dependency graph of the theory. Whenever two earlier canon fragments make the same structural claim in different language, the present version keeps one canonical form and demotes the others to historical context.

12.14 Research-track companion

The sectors that are useful for continuity but too long or too speculative for a minimal main-canon reading are also collected in the companion file `canon-0.8.0-research-track.tex`. That excerpt includes the thermodynamic-radiation interface, the minimal-coupling QED analogy, the ribbon-invariant chirality refinement, the particle-candidate mapping layer, the hydrodynamic exchange / Pauli-barrier estimate, the numerical-benchmark policy block, and the gauge-sector roadmap. The integration subsections below retain the full in-context treatments with labels and cross-references; the companion file provides a portable standalone copy. Editorial separation does not reject these sectors; it prevents the core narrative from over-claiming closure.

12.15 Integration Policy

This section records only material from prior internal SST development that can be retained inside the v0.8.0 architecture *without* weakening the epistemic classification framework of Section 4.

Rule. Older canon documents are *not* cited as external authorities. Instead, transferable content is rewritten here as current-canon statements, each labeled as **[ORTHODOX]**, **[DERIVED]**, or **[SPECULATIVE]**.

12.16 Selective Recovery from Pre-v0.7 Canon Layers

The present draft absorbs only those legacy ingredients that can be rewritten in the current epistemic format without appealing to earlier canon versions as authorities.

Analogue line-element bridge. Legacy SST drafts recorded a schematic analogue line element and a metric–clock identification of the form

$$ds^2 = -c^2\, S_{(t)}^{\circlearrowleft}(x)^2\, dt^2 + \delta_{ij}\, dx^i\, dx^j, \tag{129}$$

$$g_{00}(x) = -S_{(t)}^{\circlearrowleft}(x)^2. \tag{130}$$

Recovered legacy sectors include:

- the finite-core atomic bridge, Eqs. (79)–(81);
- the analogue line-element bridge, Eqs. (129)–(130);
- compact research-track interfaces to thermodynamic radiation and minimal-coupling QED analogies;
- topological chirality refinements based on standard ribbon invariants.

Legacy material not imported includes version history, cosmogonic narrative, and phenomenological galaxy-scale laws that are not yet sufficiently integrated with the present minimal axiom system.

12.17 Thermodynamic Radiation Interface

A recurring legacy idea is that a local swirl-energy density may define an effective temperature field and thereby an emergent radiation sector.

Let

$$T_{\text{eff}} = T_{\text{eff}}\big(\rho_E,\, \omega,\, S_{(t)}\big) \tag{131}$$

denote an unspecified constitutive relation. A minimal legacy interface then asks whether Planck-type spectra can emerge from a medium whose local excitation scale is set by T_{eff}.

[ORTHODOX]: blackbody spectra are governed by Planck's law once a valid temperature variable is defined.

[SPECULATIVE]: SST does not yet derive Eq. (131), so this sector remains a research program rather than canonically closed physics.

12.18 Minimal Coupling Interface to Quantum Electrodynamics

Early research notes repeatedly explored whether an effective vector potential could be built from circulation variables. The most conservative placeholder takes the form

$$\nabla \to \nabla - i\,\frac{m}{\hbar}\,\mathbf{A}_{\text{eff}}, \qquad \mathbf{A}_{\text{eff}} = \chi\,\mathbf{v}, \tag{132}$$

where χ is a scalar weight and $\mathbf{v}$ is the local carrier velocity field.

[ORTHODOX]: minimal coupling is the standard gateway from free to gauge-coupled wave dynamics.

[SPECULATIVE]: Eq. (132) is only an analogy template unless a full gauge-covariant derivation is supplied.

12.19 Ribbon-Invariant Chirality Refinement

The older canon layers also suggested using the Călugăreanu–White–Fuller relation

$$Lk = Tw + Wr \tag{133}$$

as a refinement of the knot-based matter/antimatter dictionary.

[ORTHODOX]: Eq. (133) is a standard geometric identity for framed curves and ribbons.

[DERIVED]: it can be used internally to separate writhe-dominated from twist-dominated realizations of the same coarse knot class.

[SPECULATIVE]: the mapping from $\operatorname{sign}(Tw)$ or related chirality measures to particle/antiparticle sectors remains a model hypothesis.

12.20 Particle-Candidate Mapping Layer

[SPECULATIVE] Working dictionary: a minimal knot / composition mapping retained from earlier SST work is summarized in Table 1. This table is organizational rather than definitive; it records the current topological assignments used by the mass and taxonomy program.

[CRITICAL NOTE] Table 1 is a working dictionary only. A full canonical assignment requires an explicit invariant set and a mass-functional comparison program.

12.21 Hydrodynamic Exchange and the Pauli Barrier

[ORTHODOX] Fluid-mechanical input: for two thin filamentary loops C_1 and C_2 in an incompressible medium, the Biot–Savart interaction energy is [6]

$$E_{\text{int}} = \frac{\rho_f \Gamma_1 \Gamma_2}{4\pi} \oint_{C_1} \oint_{C_2} \frac{d\mathbf{l}_1 \cdot d\mathbf{l}_2}{\|\mathbf{r}_1 - \mathbf{r}_2\|}. \tag{134}$$

Table 1: Minimal knot-class to particle-candidate summary retained in v0.8.0.

Knot class / composition	Candidate	Status / role
3_1	electron	baseline lepton calibration
$T(2,5)$ or equivalent higher torus class	muon	lepton hierarchy candidate
$T(2,7)$ or equivalent higher torus class	tau	lepton hierarchy candidate
$5_2 + 5_2 + 6_1$	proton	composite baryon candidate
analogous nearby composite class	neutron	composite baryon candidate

[DERIVED] SST bridge: if two identical electron-candidate loops are forced into the same spatial state, regularization at the core radius r_c yields the overlap penalty

$$V_{\text{Pauli}}(\mathbf{d}) \approx \frac{\rho_f \Gamma_0^2}{4\pi} \oint\oint \frac{d\mathbf{l}_1 \cdot d\mathbf{l}_2}{\sqrt{\|\mathbf{r}_1(\theta_1) - \mathbf{r}_2(\theta_2) + \mathbf{d}\|^2 + r_c^2}}, \tag{135}$$

with maximal-overlap scale

$$V_{\text{Pauli}}^{\max} \approx \frac{\rho_f \Gamma_0^2}{4\pi} \left(\frac{L}{r_c}\right) \mathcal{O}(1). \tag{136}$$

Using the canonical values

$$\begin{aligned} \rho_f &= 7.0 \times 10^{-7}\ \mathrm{kg\,m^{-3}}, \\ r_c &= 1.40897017 \times 10^{-15}\ \mathrm{m}, \\ \Gamma_0 &= 2\pi r_c \|\mathbf{v}_{\circlearrowleft}\| \approx 9.6836 \times 10^{-9}\ \mathrm{m^2\,s^{-1}}, \end{aligned} \tag{137}$$

and $L \sim 2\pi a_0$ at the hydrogenic scale, one obtains the benchmark estimate

$$V_{\text{Pauli}}^{\max} \sim 1.23 \times 10^{-18}\ \mathrm{J} \sim 7.6\ \mathrm{eV}. \tag{138}$$

[CRITICAL NOTE] This is retained as a canonical scale benchmark, not yet as a closed theorem. Its role is to show that the exclusion-energy estimate falls in an atomically relevant window rather than at obviously pathological scales.

As written, Eq. (138) should be interpreted as an order-of-magnitude benchmark attached to the canonical constants, not yet as a unique parameter-free prediction.

12.22 Numerical Benchmark Layer

[DERIVED] Computational-program requirement: earlier SST work contained explicit benchmark tables for particle and atomic masses. In v0.8.0, the benchmark layer is retained as a reproducibility requirement rather than as a standalone authority claim.

Table 2: Benchmark summary structure to be populated from canonical scripts and archived CSV output.

Particle	m_{exp} (MeV)	m_{SST} (MeV)	rel. error	mode	class
e	0.51099895	0.51099895	0	predictive/closure	3_1
μ	105.6583745	...	...	predictive	candidate torus class
τ	1776.86	...	...	predictive	candidate torus class
p	938.272088	...	...	predictive or closure	composite class
n	939.565420	...	...	predictive or closure	composite class

[CRITICAL NOTE] A future v0.8.x benchmark appendix should include:

- the exact script or package path used for generation,

- archived CSV outputs,
- mode definitions (predictive vs. exact-closure),
- uncertainty propagation on canonical constants and geometric factors.

This requirement is included to keep the benchmark layer falsifiable and reproducible, rather than merely illustrative.

12.23 Gauge-Sector Roadmap

[SPECULATIVE] Structural roadmap: a retained minimal statement of the gauge program is:

$$\mathfrak{g}_{\text{eff}} = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1), \tag{139}$$

with effective gauge phases organized as holonomy data of multi-director transport.

A minimal phase-based representation is

$$A_\mu^a(x) = \partial_\mu \theta^a(x), \tag{140}$$

and clock coupling acts as a common phase reference,

$$\theta^a(x) \mapsto \theta^a(x) + q^a \chi(x) \quad \Rightarrow \quad A_\mu^a \mapsto A_\mu^a + q^a \partial_\mu \chi. \tag{141}$$

[CRITICAL NOTE] The gauge layer is retained in v0.8.0 only as a roadmap statement. Running couplings, anomaly cancellation, and realistic mixing matrices remain open work.

13 Discussion and Limits

13.1 Status of the recovered v0.5.12 sectors

The material reintroduced here does not have uniform epistemic status.

- The Chronos–Kelvin and pressure-law sectors are orthodox fluid-mechanical structures with SST reinterpretations.
- The analogue-metric and clock-field EFT layers are orthodox comparison tools, but only partially derived within SST.
- The R/T measurement bridge and weak-mixing stiffness ansatz remain explicitly speculative.

13.2 What v0.8.0 now recovers from v0.5.12

After the present patch, v0.8.0 no longer omits the main v0.5.12 kernel sectors that are structurally compatible with the current architecture:

- Chronos–Kelvin / clock transport,
- zero-parameter reparameterization discipline,
- Swirl–EM and photon / unknot bridge,
- swirl pressure law,
- analogue metric and clock-field gravity bridge,
- R/T measurement bridge,
- weak-mixing bridge ansatz.

13.3 Open items that remain outside the present closure

The following remain open and are *not* upgraded to full canonical closure in this draft:

- a first-principles derivation of the full SM gauge sector;
- a completed EWSB derivation tied to the canon constants;
- a rigorous derivation of the R/T measurement kernel from continuum dynamics;
- a unique dark-sector taxonomy with experimentally anchored observables.

13.4 What is established at canon level

The present document supports the following as canon-level structures:

- the primitive constant chain $(\rho_f, v_{\circlearrowleft}, \omega_c) \to r_c \to \Gamma_0$,
- the Swirl-Clock as the coarse-grained relational time factor,
- delay-induced mode selection as the preferred discreteness mechanism,
- the atomic bridge and spectroscopic sectors as compatibility filters,
- the topology-first program for mass organization.

These claims are not equally proven, but they are the sectors on which the rest of the canon is structurally built.

13.5 What remains closure-level or conjectural

Several central SST ambitions remain open:

1. a rigorous derivation of full particle spectra from topology alone,
2. a first-principles derivation of charge assignments and gauge couplings,
3. a complete many-body derivation of shell filling and exclusion,
4. a full demonstration that delay plus topology reproduces the observed quantum rule set without importing it.

The canon records these as active programs rather than as settled conclusions.

13.6 Compatibility limits

Any viable SST realization must remain compatible with:

- atomic spectroscopy,
- relativistic effective kinematics,
- precision electroweak data where an SST bridge is claimed,
- the absence of large low-energy deviations from standard quantum mechanics.

This is why the canon repeatedly imposes suppression, weak-coupling, and low-energy recovery conditions.

13.7 Minimal falsifiers

The present canon would be materially weakened by any of the following:

- failure of the constant chain to remain dimensionally and numerically self-consistent,
- absence of any workable suppression mechanism for internal modes at atomic energies,
- inability to formulate a non-circular route from delay dynamics to discrete stable spectra,
- systematic contradiction between any SST atomic bridge and precision spectroscopy.

Conversely, strong support would come from a successful theorem ladder showing how discrete spectra, topological stabilization, and low-energy effective quantum structure arise from one shared substrate model.

13.8 Methodological limit

The canon is deliberately hybrid in style. Some sectors are written as exact closures, some as effective models, and some as programmatic roadmaps. This is a feature rather than a flaw, provided the epistemic status of each claim remains explicit. The document should therefore not be judged as though every section were intended to be a stand-alone journal theorem.

13.9 Near-term priorities

The next canon-strengthening tasks are clear:

1. complete the theorem ladder for the mass and clock sectors,
2. expand numerical validation blocks using the canonical constants,
3. refine the atomic bridge into sharper benchmark observables,
4. decide which optional sectors belong in the core canon and which should migrate to separate technical notes or the research-track companion.

13.10 Final limit statement

SST is presently best understood as a structured unification program with several strong internal bridges and several open derivational gaps. The canon is therefore mature enough to organize the theory coherently, but not yet so mature that every phenomenological sector should be presented as theorem-complete.

13.11 Status of Legacy Recoveries

The additional material imported here from older canon layers does not have uniform status.

- The analogue-metric bridge and ribbon identity are mathematically orthodox tools.
- Their SST interpretation is partially derived but not fully closed dynamically.
- The thermal-radiation and QED-interface sectors remain explicitly research-track.

This section therefore serves as a controlled retention layer, not as a blanket validation of all legacy material.

Accordingly, these recoveries should be read as structured extensions of the minimal core, not as fully promoted final doctrine.

13.12 What v0.8.0 Now Retains

After integration, the present Canon retains the following prior-development sectors in current-canon form:

- a particle-candidate topological dictionary (Section 12.20);
- a hydrodynamic exchange-energy estimate for fermionic exclusion (Section 12.21);
- a benchmark / reproducibility layer (Section 12.22);
- a minimal gauge-sector roadmap (Section 12.23);
- a clock-field EFT bridge and Poisson gravity limit (Section 10).

13.13 What is Deliberately Not Claimed

The present document still does *not* claim:

- a completed first-principles derivation of the Standard Model;
- a unique finalized knot-to-particle dictionary;
- a closed proof of the exclusion barrier beyond the current scaling benchmark;
- a fully renormalized relativistic field theory for the gauge and clock sectors.

These omissions are deliberate. v0.8.0 is intended to remain logically strict even when phenomenological sectors are reintroduced.

13.14 Canon Rule Going Forward

Any further reintegrated material must satisfy all of the following:

1. it can be written without citing older internal canon versions as authorities;
2. it can be assigned an explicit epistemic label;
3. it includes either an internal derivation path or a reproducible benchmark route;
4. it preserves compatibility with the orthodox limits recorded earlier in the document.

A Full Derivations

A.1 Canonical constant chain

The present canon uses the closure chain

$$r_e = \frac{\alpha\hbar}{m_e c}, \tag{142}$$

$$r_c = \frac{r_e}{2}, \tag{143}$$

$$\omega_c = \frac{m_e c^2}{\hbar}, \tag{144}$$

$$v_{\circlearrowleft} = \omega_c r_c, \tag{145}$$

$$\Gamma_0 = 2\pi r_c v_{\circlearrowleft}. \tag{146}$$

Substituting $r_c = \alpha\hbar/(2m_e c)$ into $v_{\circlearrowleft} = \omega_c r_c$ gives

$$v_{\circlearrowleft} = \left(\frac{m_e c^2}{\hbar}\right)\left(\frac{\alpha\hbar}{2m_e c}\right) = \frac{\alpha}{2}c, \tag{147}$$

hence the canonical ratio

$$\eta_0 = \frac{v_{\circlearrowleft}}{c} = \frac{\alpha}{2}. \tag{148}$$

A.2 Core density closure

Writing the electron rest energy as localized swirl energy inside a core region leads to the closure

$$\rho_{\text{core}} \sim \frac{m_e c^2}{2\pi v_{\circlearrowleft}^2 r_c^3}, \tag{149}$$

which is the form already recorded in Eq. (17). This relation is dimensionally exact:

$$\left[\frac{m_e c^2}{v^2 r^3}\right] = \frac{\mathrm{kg\,m^2\,s^{-2}}}{\mathrm{m^2\,s^{-2}\,m^3}} = \mathrm{kg\,m^{-3}}. \tag{150}$$

A.3 Geometric gate identity

Using the Compton wavelength $\lambda_c = h/(m_e c)$ and the core closure $r_c = \alpha\hbar/(2m_e c)$, one obtains

$$\frac{\lambda_c}{\pi r_c} = \frac{h/(m_e c)}{\pi\alpha\hbar/(2m_e c)} = \frac{2h}{\pi\alpha\hbar} = \frac{4}{\alpha}. \tag{151}$$

This is the canonical geometric shielding factor used throughout the late mass-functional papers.

A.4 Useful numerical anchors

With the canonical values

$$v_{\circlearrowleft} \approx 1.09384563 \times 10^6 \ \mathrm{m\,s^{-1}}, \tag{152}$$

$$r_c \approx 1.40897017 \times 10^{-15} \ \mathrm{m}, \tag{153}$$

$$\Gamma_0 \approx 9.6836 \times 10^{-9} \ \mathrm{m^2\,s^{-1}}, \tag{154}$$

one has the Compton turnover time

$$\tau_c = \frac{2\pi r_c}{\|\mathbf{v}_{\circlearrowleft}\|} = \frac{2\pi}{\omega_c}, \tag{155}$$

and the geometric gate

$$\frac{\lambda_c}{\pi r_c} \approx 5.48 \times 10^2. \tag{156}$$

B Delay Mathematics

B.1 Fixed-point condition for locked modes

Starting from the delayed phase equation

$$\dot{\phi}(t) = \omega_0 + \kappa \sin\big(\phi(t-\tau) - \phi(t)\big), \tag{157}$$

seek a rigidly rotating solution

$$\phi(t) = \Omega t + \phi_\star. \tag{158}$$

Substitution gives the algebraic mode condition

$$\Omega = \omega_0 + \kappa\sin(\Omega\tau), \tag{159}$$

which admits multiple branches labelled by the effective phase winding across the delay interval.

B.2 Linear stability of a locked branch

Let

$$\phi(t) = \Omega t + \epsilon(t), \qquad |\epsilon| \ll 1. \tag{160}$$

Linearizing yields

$$\dot{\epsilon}(t) = \kappa \cos(\Omega\tau)\big(\epsilon(t-\tau) - \epsilon(t)\big). \tag{161}$$

Using the exponential ansatz $\epsilon(t) \propto e^{\lambda t}$ gives the characteristic equation

$$\lambda = \kappa \cos(\Omega\tau) \left(e^{-\lambda\tau} - 1\right). \tag{162}$$

Stability requires $\mathrm{Re}(\lambda) < 0$ for all nontrivial roots. Thus discrete spectra in SST are not arbitrary: only those branches satisfying both the fixed-point equation and the stability inequality survive dynamically.

B.3 Small-delay expansion

For $|\lambda\tau| \ll 1$,

$$e^{-\lambda\tau} = 1 - \lambda\tau + \mathcal{O}((\lambda\tau)^2), \tag{163}$$

so the characteristic equation becomes

$$\lambda \approx -\kappa\tau \cos(\Omega\tau)\, \lambda. \tag{164}$$

Nontrivial damping therefore requires the effective factor $\kappa\tau\cos(\Omega\tau)$ to have the correct sign. This provides a simple branch-selection criterion at weak delay.

B.4 Interpretive consequence

The important canon point is that discreteness is produced in two stages:

$$\text{delay} \;\longrightarrow\; \text{multiple algebraic branches} \;\longrightarrow\; \text{stability filter.} \tag{165}$$

Quantization is therefore not inserted by hand; it is the residue of a dynamical selection problem on a delayed closed carrier.

C Conversation-Derived Insights

C.1 Role of this appendix

This appendix records recurrent insights that emerged during the project-level development of the canon and are useful for guiding future work, even when they do not yet qualify as theorem-level results.

C.2 Topology first, layers second

A central refinement of the recent canon is the separation

$$\text{particle identity} \;\longleftrightarrow\; \text{topology}, \qquad \text{excitation or dressing state} \;\longleftrightarrow\; \text{layer index.} \tag{166}$$

This prevents Golden-layer fits from being mistaken for a species ontology and keeps the topology program structurally cleaner.

C.3 Spectral origin of $\zeta(3)$

The project discussions repeatedly converged on the conclusion already encoded in Section 5.2: the appearance of $\zeta(3)$ is best interpreted as a spectral invariant of a cubic-weighted closed-carrier mode tower, not as a direct invariant of threefold geometry by itself.

C.4 Atomic bridge status discipline

Another recurring project insight is that the atomic bridge is strongest when treated as a benchmark layer with explicit status tags. Hydrogenic consistency, branch-sensitive couplings, and the spectroscopic gap form a robust bridge; shell filling, exclusion, and full heavy-atom structure remain open programs rather than completed derivations.

C.5 Master-equation viewpoint

A major organizational gain of the recent conversations is the recognition that many SST sectors can be read as different projections of one shared structure:

$$\text{medium scale} \times \text{topology} \times \text{geometry} \times \text{clocking}. \tag{167}$$

Even when this is not written as a single canonical equation inside the present draft, it remains a useful guide for reducing duplication across the mass, time, and thermodynamic sectors.

C.6 Dual-vacuum and photon sector status

The project-level discussions also clarified a canon policy: the photon/torsion bridge and the dual-vacuum phenomenology are valuable and fertile, but they should not silently override the simpler core canon. They are best treated as phenomenological extensions built on top of the primitive constant chain and Swirl-Clock structure.

C.7 Open theorem ladder

The most important unfinished task identified across the project is not adding new sectors, but tightening the derivational ladder between existing ones. In practice this means:

1. turning heuristic closures into explicit lemmas,
2. attaching numerical validation to each nontrivial scaling law,
3. separating benchmark results from full ontological claims.

This appendix therefore serves as a reminder that canon quality increases more by sharpening structure than by increasing conceptual breadth.

References

[1] J. K. Hale and S. M. Verduyn Lunel, *Introduction to Functional Differential Equations*, Springer (1993).

[2] CMS Collaboration, High-precision measurement of the W boson mass with the CMS experiment, Nature **652** (2026), 321–337, DOI: 10.1038/s41586-026-10168-5.

[3] S. Navas et al. (Particle Data Group), Review of Particle Physics, Phys. Rev. D **110** (2024), 030001, DOI: 10.1103/PhysRevD.110.030001.

[4] J. de Blas et al., Global analysis of electroweak data in the Standard Model, Phys. Rev. D **106** (2022), 033003, DOI: 10.1103/PhysRevD.106.033003.

[5] CDF Collaboration, High-precision measurement of the W boson mass with the CDF II detector, Science **376** (2022), 170–176, DOI: 10.1126/science.abk1781.

[6] G. K. Batchelor, *An Introduction to Fluid Dynamics*, Cambridge University Press (1967).

[7] T. Jacobson, Einstein-Æther gravity: a status report, PoS QG-PH (2007) 020, DOI: 10.22323/1.043.0020.

[8] D. Blas, O. Pujolàs, and S. Sibiryakov, Models of non-relativistic quantum gravity: the good, the bad and the healthy, JHEP **04** (2011) 018, DOI: 10.1007/JHEP04(2011)018.

[9] B. P. Abbott et al. (LIGO Scientific Collaboration and Virgo Collaboration), GW170817: Observation of gravitational waves from a binary neutron star inspiral, Phys. Rev. Lett. **119** (2017) 161101, DOI: 10.1103/PhysRevLett.119.161101.

[10] W. G. Unruh, Experimental black-hole evaporation?, Phys. Rev. Lett. **46** (1981), 1351–1353, DOI: 10.1103/PhysRevLett.46.1351.

[11] M. Visser, Acoustic black holes: horizons, ergospheres and Hawking radiation, Class. Quantum Grav. **15** (1998), 1767–1791, DOI: 10.1088/0264-9381/15/6/024.

[12] P. Painlevé, La mécanique classique et la théorie de la relativité, C. R. Acad. Sci. (Paris) **173** (1921), 677–680.

[13] A. Gullstrand, Allgemeine Lösung des statischen Einkörperproblems in der Einsteinschen Gravitationstheorie, Arkiv. Mat. Astron. Fys. **16**(8) (1922), 1–15.

[14] E. Schrödinger, Quantisierung als Eigenwertproblem, Annalen der Physik **384** (1926), 361–376, DOI: 10.1002/andp.19263840404.

[15] J. D. Jackson, *Classical Electrodynamics*, 3rd ed., Wiley (1999).

[16] M. Planck, Ueber das Gesetz der Energieverteilung im Normalspectrum, Annalen der Physik **309** (1901), 553–563, DOI: 10.1002/andp.19013090310.

[17] W. Wien, Ueber die Energieverteilung im Emissionsspectrum eines schwarzen Körpers, Annalen der Physik **294** (1896), 662–669, DOI: 10.1002/andp.18962940310.

[18] G. Călugăreanu, Sur les classes d'isotopie des noeuds tridimensionnels et leurs invariants, Czech. Math. J. **11** (1961), 588–625.

[19] J. H. White, Self-linking and the Gauss integral in higher dimensions, Amer. J. Math. **91** (1969), 693–728, DOI: 10.2307/2373348.

[20] P. G. Saffman, *Vortex Dynamics*, Cambridge University Press (1992).

[21] H. K. Moffatt, The degree of knottedness of tangled vortex lines, J. Fluid Mech. **35** (1969), 117–129, DOI: 10.1017/S0022112069000991.

[22] W. H. Zurek, Decoherence, einselection, and the quantum origins of the classical, Rev. Mod. Phys. **75** (2003), 715–775, DOI: 10.1103/RevModPhys.75.715.

[23] S. Weinberg, A model of leptons, Phys. Rev. Lett. **19** (1967), 1264–1266, DOI: 10.1103/PhysRevLett.19.1264.